AI Quality Inspection System for Packaged Food Production Lines

An end-to-end computer vision pipeline for real-time quality inspection of bottles on a conveyor belt. The system detects cap defects and fill level anomalies using YOLO-based object detection classification, and surfaces results through a live monitoring dashboard.

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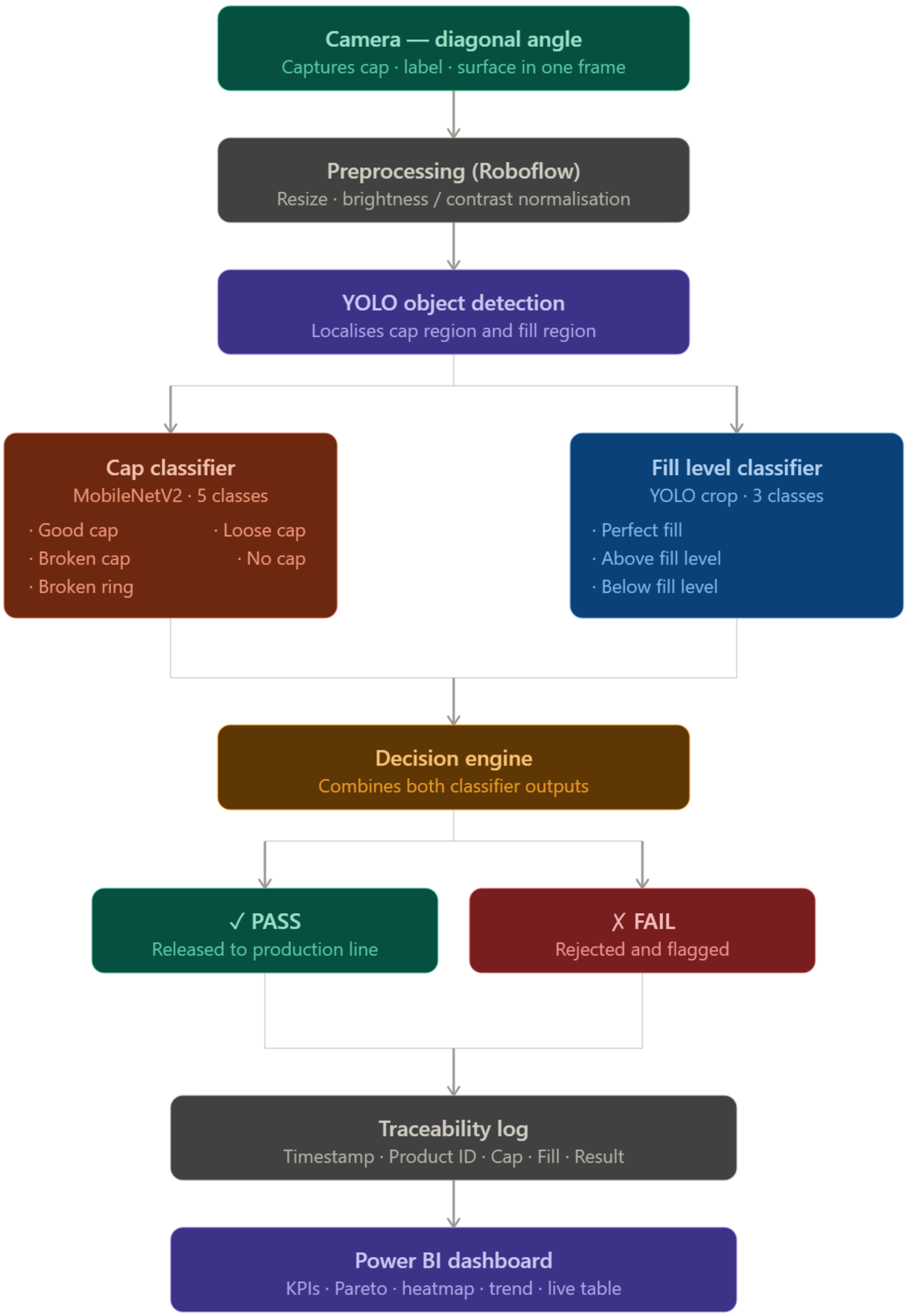
Overview

Manual quality inspection on high-throughput production lines is error-prone and slow. This system replaces manual spot-checks with a continuous, automated vision pipeline that inspects every bottle in real time and flags defects before they reach customers.

Inspection targets:

- Cap condition (5 classes)
- Overall bottle disposition: `PASS` or `FAIL`

System Architecture



Camera Setup

The camera is positioned **diagonally** to the conveyor belt, capturing a combined view of the cap, label, and bottle surface in a single frame.

Benefits of diagonal positioning:

- Covers cap, label, and surface defects simultaneously
- Eliminates the need for multiple cameras at separate stations
- Reduces hardware cost and system complexity while maintaining defect coverage

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Image Preprocessing

All captured frames are preprocessed before inference using **Roboflow**:

Step	**Description**
Resize	Standardise image dimensions for model input
Lighting adjustment	Normalise brightness and contrast for consistent inference under variable line lighting

Detection & Classification

Cap Inspection

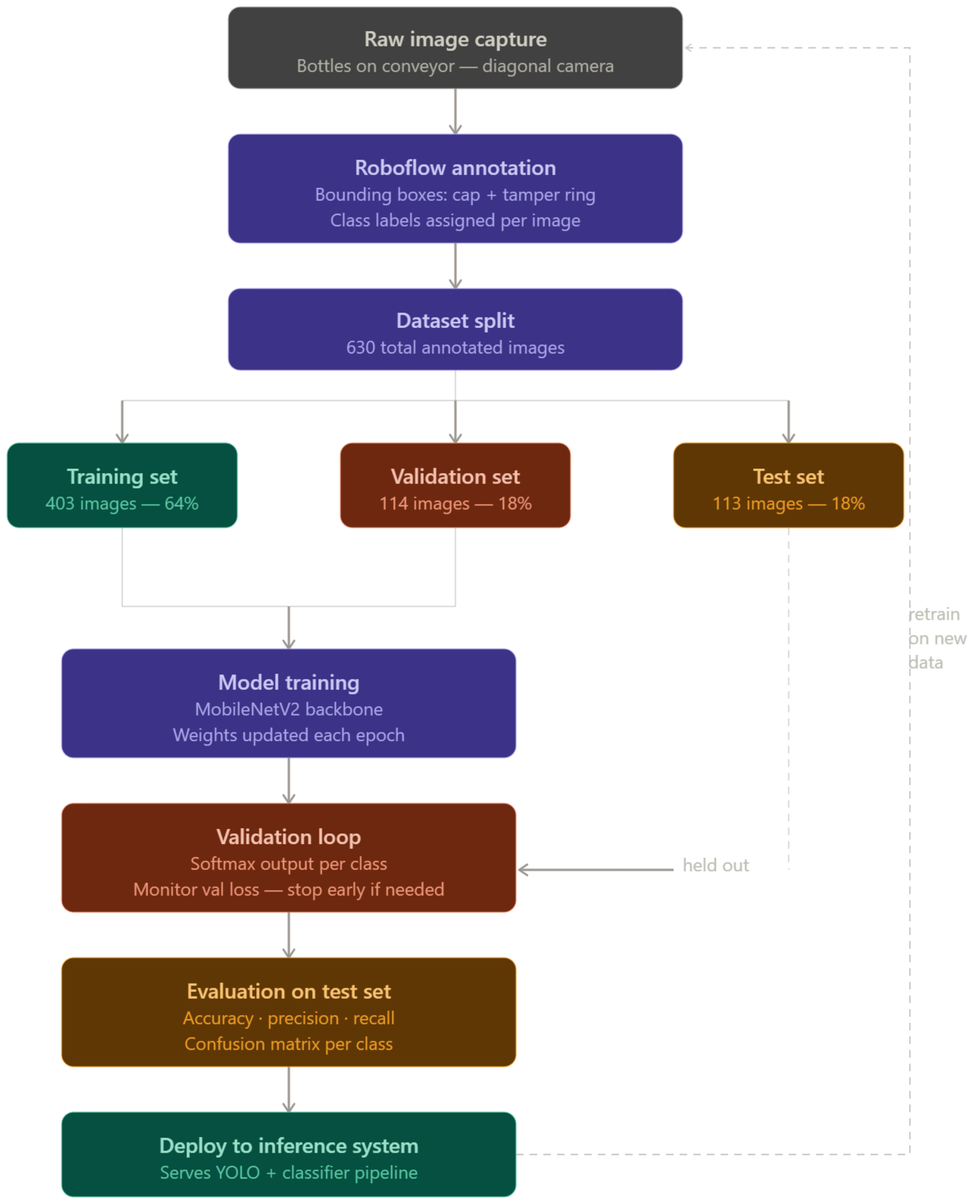
Model: YOLO8
Detection tool: YOLO (localises the cap region before classification)

The classifier distinguishes five cap states:

Class	Description
`Good Cap`	Cap properly sealed and intact
`Broken Cap`	Cap physically damaged or cracked
`Broken Ring`	Safety ring detached or broken
`Loose Cap`	Cap present but not fully tightened
`No Cap`	Bottle missing the cap entirely

Annotation approach: Bounding boxes drawn around the cap and tamper-evident ring.

Training Pipeline



Decision Engine

After both classifiers run, a lightweight decision engine combines the results:

Cap Status	Fill Status	Final Result
Good Cap	Perfect Fill	PASS
Loose Cap	Perfect Fill	FAIL
Good Cap	Underfill	FAIL
Any defect class	Any	FAIL

Results are written to a timestamped log with `Product ID`, `Cap Status`, `Fill Status`, and `Result` for full traceability.

Dashboard

The monitoring dashboard answers five operational questions at a glance:

1. How many bottles / packages are good?
2. What defects are occurring?
3. Where are defects increasing?
4. Is the line meeting quality targets?
5. What is the trend over time?

Executive KPI Row

KPI	Description	Example
Total Inspected	Total packages checked	25,486
Good Packages	Passed inspection	24,912
Rejected Packages	Failed inspection	574
Yield (%)	Good / Total × 100	97.75%

KPI	Description	Example
Defect Rate (%)	Rejected / Total × 100	2.25%
Current Line Status	Running / Stopped	Running

Cap Inspection Panel

Visual: **Donut chart** — distribution across the five cap classes.

Metric	Observation	Recommended Action
Broken Cap ↑	Machine overtightening caps	Recalibrate torque control
Loose Cap (highest defect)	Spindle wear or torque inconsistency	Inspect capping unit; replace worn parts
No Cap ↑	Feeder / sensor malfunction	Inspect cap feeder and sensor health
Total Cap Defects > 4%	Capping station degrading	Prioritise preventive maintenance

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Fill Level Inspection Panel

Visual: **Pie chart** — distribution across fill classes.

Metric	Observation	Recommended Action
Above Fill ↑	Filling valve drifting high	Recalibrate filling machine and verify valve timing
Below Fill ↑	Nozzle blockage or pump wear	Inspect pumps, nozzles, and flow control
Rejected ↑	Severe filling deviation	Immediate root-cause investigation
Total Fill Defects > 5%	Filling station needs attention	Focus improvement on filling process

Defect Pareto Chart

Visual: **Bar chart** — ranked by defect count to guide maintenance prioritisation.

Defect Type	Count
Loose Cap	210
Underfill	170
No Cap	95
Broken Cap	54
Overfill	45

Real-Time Production Trend

Visual: **Line chart** — bottles inspected per hour.

Time	Bottles Inspected
9 AM	1,200
10 AM	1,350
11 AM	1,280

Quality Trend Over Time

Visual: **Line chart** — good vs. rejected unit percentage over time. Helps identify machine drift early.

Time	Good Units (%)
9 AM	99.2%
10 AM	98.5%

Time	Good Units (%)
11 AM	97.8%

Defects by Hour Heatmap

Quickly highlights the most problematic production periods.

Hour	Rejections
9 AM	12
10 AM	18
11 AM	35

Camera Detection Summary Table

Per-bottle traceability log for audit and root-cause analysis:

Timestamp	Cap Status	Fill Status	Result
10:15:21	Good Cap	Perfect Fill	PASS
10:15:23	Loose Cap	Perfect Fill	FAIL
10:15:25	Good Cap	Underfill	FAIL

Data Model

Each inspection event is logged with the following schema (used as the Power BI data source):

Field	Type	Example
`Timestamp`	datetime	`10:15:21`
`Product ID`	string	`P001`

Field	Type	Example
`Cap Status`	categorical	`Good Cap`
`Fill Status`	categorical	`Perfect Fill`
`Result`	binary	`Pass` / `Fail`

Tech Stack

Component	Technology
Object detection	YOLO
Classification backbone	MobileNetV2
Annotation & dataset management	Roboflow
Inference runtime	Python
Dashboard & visualisation	Power BI

Sample Results

[10:15:21] P001 | Cap: Good Cap | Fill: Perfect Fill | → PASS

[10:15:23] P002 | Cap: Loose Cap | Fill: Perfect Fill | → FAIL

[10:15:25] P003 | Cap: Good Cap | Fill: Underfill | → FAIL

Built for automated quality assurance on high-throughput packaged food production lines.